

SOLENZARA AB, France

WMO: 077650

Lat: 41.9239N

Lon: 9.3972E

Elev: 56

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	35.3	37.8	18.7	14.1	44.6	22.3	16.8	45.2	25.1	52.7	22.5	53.2	7.9	270	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	88.2	72.2	86.3	72.5	84.8	72.1	76.6	84.8	75.4	83.7	74.2	82.7	8.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	126.0	83.4	72.4	120.1	82.5	71.0	114.5	81.5	40.0	85.0	38.8	83.8	37.7	82.8	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	17.4	14.5	32.0	94.8	2.6	2.9	30.2	96.9	28.7	98.6	27.2	100.3	25.4	102.4
			28.6	78.1	2.3	1.4	26.9	79.1	25.6	79.9	24.3	80.6	22.7	81.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	49.6	49.4	53.1	57.5	63.9	71.1	76.4	76.7	71.0	64.4	56.5	51.5	(6)
	DBStd	10.64	4.40	4.56	4.29	3.84	3.68	4.42	3.49	3.23	3.94	4.35	4.95	4.90	(7)
	HDD50	184	58	57	19	1	0	0	0	0	0	0	9	40	(8)
	HDD65	2320	478	437	369	226	65	6	1	0	4	60	256	418	(9)
	CDD50	4506	46	40	116	226	432	632	817	828	629	448	205	87	(10)
	CDD65	1166	0	0	1	1	31	189	353	363	183	43	2	0	(11)
	CDH74	9176	0	1	5	1	150	1347	3285	3279	1008	99	1	0	(12)
	CDH80	2395	0	0	1	0	8	253	988	1004	136	5	0	0	(13)
(14) Wind	WSAvg	6.8	7.0	7.2	7.4	6.7	6.3	6.2	6.6	6.5	7.0	6.5	6.8	7.2	(14)
(15) Precipitation	PrecAvg	30.0	2.8	2.2	2.8	2.8	1.9	1.0	0.4	0.8	3.1	4.1	4.2	3.8	(15)
	PrecMax	45.6	6.9	7.5	9.6	5.6	9.0	3.0	1.3	3.6	12.5	10.4	12.5	9.2	(16)
	PrecMin	20.6	0.0	0.3	0.0	0.5	0.3	0.0	0.0	0.0	0.6	0.1	0.8	0.0	(17)
	PrecStd	6.3	1.7	1.8	2.6	1.5	1.8	0.8	0.4	0.9	2.6	2.5	2.2	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.6	65.5	71.3	73.2	81.0	88.3	91.2	91.9	85.8	79.9	72.3	68.3	(19)
		MCWB	53.3	52.7	55.9	58.7	64.4	69.9	68.0	70.9	68.8	65.1	61.7	55.3	(20)
	2%	DB	62.1	62.0	66.0	70.4	78.1	85.4	88.2	88.3	83.1	76.6	68.8	64.3	(21)
		MCWB	51.8	51.2	53.7	58.5	63.7	71.5	72.5	72.5	69.6	66.3	59.9	54.4	(22)
	5%	DB	59.3	59.6	63.2	68.0	75.2	82.9	86.5	86.3	81.4	74.3	66.8	61.8	(23)
		MCWB	51.6	51.2	53.5	57.4	62.9	70.2	72.8	72.9	69.1	65.5	59.2	54.0	(24)
	10%	DB	57.4	57.9	61.3	66.0	72.9	80.8	84.9	84.8	79.7	72.3	64.7	59.5	(25)
		MCWB	50.9	50.6	53.2	57.0	62.4	69.1	72.4	72.3	68.1	64.2	58.1	52.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	56.2	58.7	62.8	69.2	75.6	78.3	78.1	75.4	71.1	65.4	59.4	(27)
		MCDB	61.0	60.6	65.3	69.4	76.8	84.1	86.5	85.9	81.9	75.7	68.8	63.5	(28)
	2%	WB	54.9	54.4	57.0	60.8	66.7	73.9	76.6	76.8	73.2	69.2	62.9	57.4	(29)
		MCDB	59.2	59.2	62.3	67.3	74.3	82.5	85.1	84.9	80.1	73.9	66.7	61.6	(30)
	5%	WB	53.3	53.0	55.8	59.5	65.3	72.0	75.3	75.6	71.6	67.5	61.0	55.6	(31)
		MCDB	57.8	57.8	60.8	65.7	72.7	80.7	83.9	83.8	79.1	72.5	64.9	59.8	(32)
	10%	WB	51.8	51.6	54.7	58.3	63.8	70.4	74.2	74.4	70.0	65.9	59.2	54.0	(33)
		MCDB	56.5	56.6	59.8	64.1	70.9	78.9	82.9	83.0	77.2	71.1	63.6	58.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	14.3	14.6	15.0	15.4	16.2	16.1	15.9	15.0	14.0	13.2	12.6	(35)
		MCDBR	14.9	16.2	17.2	16.9	18.3	17.6	17.6	17.5	16.4	15.3	14.5	13.9	(36)
	5% WB	MCWBR	10.6	11.3	11.3	10.8	11.3	11.4	12.0	12.7	11.4	10.7	9.5	10.0	(37)
		MCWBR	12.7	14.7	15.1	15.1	16.1	16.0	16.0	14.9	13.3	12.4	12.4	12.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.333	0.351	0.382	0.420	0.432	0.420	0.420	0.417	0.409	0.389	0.366	0.336	(40)
		taud	2.493	2.430	2.353	2.249	2.258	2.336	2.340	2.359	2.375	2.418	2.424	2.488	(41)
	Ebn at Noon		258	270	273	270	269	272	270	268	261	252	241	246	(42)
		Edh at Noon	24	29	35	42	43	40	39	37	34	29	25	22	(43)
(44) All-Sky Solar Radiation	RadAvg		555	838	1234	1645	1980	2253	2293	1991	1498	1014	625	495	(44)
	RadStd		53	80	121	100	146	108	89	97	68	85	53	52	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	N/A

Nomenclature: See separate page